GRADUATION PROJECT DEFENSE

Syrian Private University | SPU



Autonomous Vision-Guided Robotic Manipulation System for Intelligent Jigsaw Puzzle Assembly

A 4-DOF Robotic Arm System Integrated with Eye-in-Hand Computer Vision, ROS 2 Humble & Gazebo Simulation.

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SYSTEM OVERVIEW

Project Scope & Domain

4-DOF Kinematic Control

Optimized 4-DOF manipulator design built specifically for 2D planar pick-and-place tasks with reduced computational cost.

Eye-in-Hand Vision System

End-effector mounted camera providing dynamic spatial awareness and precise coordinate extraction using OpenCV.

ROS 2 & Gazebo Classic

High-fidelity physical simulation environment integrating MoveIt 2 for motion planning and inverse kinematics.

COMPARATIVE ANALYSIS

Literature Review & Benchmarking

Reference Paper	Methodology	Strengths	Limitations	Our Advantage
Felzenszwalb & Huttenlocher (2004)	Graph-based image segmentation.	Strong theoretical solver.	No physical or robotic manipulation integration.	Full physical simulation & robotic execution in ROS 2.
Guo et al. (2018)	Eye-in-Hand vision with 6-DOF arm.	High precision assembly.	6-DOF is computationally heavy & over-engineered for 2D.	Optimized 4-DOF system tailored for planar puzzle tasks.
Sharma et al. (2023)	4-DOF arm with Open-Loop vision.	Simple 4-DOF hardware structure.	Open-loop control lacks dynamic error correction.	MoveIt 2 active feedback with collision-free path planning.
Our System (SPU 2026)	**Closed-Loop Eye-in-Hand + 4-DOF IK**	**Low-cost, high precision.**	**-**	**Integrated ROS 2, MoveIt 2 & Gazebo Pipeline.**

MOTIVATION

Problem Statement vs. Proposed Solution

The Problem

- **6-DOF Over-engineering:** Traditional industrial solutions utilize 6-DOF arms for 2D tasks, causing extreme computational overhead.
- **Open-Loop Failures:** Static camera setups fail to adapt to spatial offsets or slight piece misalignments during operation.

Our Solution

- **Targeted 4-DOF Design:** Streamlined kinematic structure optimized specifically for planar assembly.
- **Eye-in-Hand Feedback:** Real-time closed-loop visual control using ROS 2, OpenCV, and Gazebo for accurate positioning.

KEY MILESTONES

Core Project Objectives

Cost-Effective 4-DOF Architecture

Eliminate unnecessary degrees of freedom for planar assembly to lower hardware overhead.

Autonomous Visual Pipeline

Extract puzzle contours, centroid pixel coordinates, and orientation via OpenCV.

Precise Spatial Mapping

Transform camera pixel frame coordinates into 3D Gazebo world coordinates seamless execution.

MoveIt 2 Trajectory Planning

Generate collision-free smooth trajectories using MoveIt 2 Inverse Kinematics solvers.

Key Contributions

Main Bullet Points:

1- From Abstract Image Matching to Physical Simulation

Bridges the gap between theoretical 2D image matching (Felzenszwalb & Huttenlocher, 2004) and physical execution by deploying a 4-DOF manipulator in Gazebo with realistic physics (gravity, friction, and collisions).

2- Next-Gen Middleware & Dynamic Trajectory Planning

Replaces legacy, static frameworks (Guo et al., 2018) with **ROS 2 Humble** and **MoveIt 2**, enabling real-time adaptive motion planning and dynamic object spawning.

3- Closed-Loop Eye-in-Hand Visual Feedback

Advances beyond open-loop pick-and-place systems (Sharma et al., 2023) by implementing continuous Eye-in-Hand visual feedback via OpenCV for real-time feature detection and 3D pose estimation (X, Y, Z, θ).



EXECUTION FLOW

Development Methodology To Simulation



NEXT STEPS

Upcoming Development Milestones

MoveIt Setup Assistant

Finalize the arm MoveIt configuration package to generate kinematic groups and end-effector definitions.

Inverse Kinematics Integration

Implement precise spatial coordinate targeting using MoveIt 2 IK solvers for smooth motion trajectory.

Gripper Mechanism Activation

Integrate a vacuum gripper plugin in Gazebo for physical attachment and placement of puzzle pieces.

Full End-to-End Pipeline

Execute complete automated loop: Detection -> MoveIt Planning -> Gripping -> Assembly Placement.

Thank You!

Questions & Academic Discussion

Autonomous Vision-Guided Robotic Manipulation System

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